TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

Bachelor of Artificial Intelligence and Robotics Engineering

AI.107

Data Structures and Algorithms

COURSE OVERVIEW

Organizing data and analyzing algorithms: arrays, lists, trees, graphs, sorting, searching, and complexity.

1

SESSION

60 MINUTES

ORIENTATION: DATA + ALGORITHMS

KEY TOPICS

- Why data structures matter
- Abstract data types
- Algorithm analysis motivation
- Course scope

TEACHING FOCUS

Frame computing as data organization plus stepwise logic.

KEY TAKEAWAY The right structure makes the algorithm efficient.

2

SESSION

60 MINUTES

COMPLEXITY ANALYSIS

KEY TOPICS

- Big O, Big Omega, Big Theta
- Time vs. space complexity
- Analyzing loops and recursion
- Best/average/worst case

TEACHING FOCUS

Quantify algorithm cost before coding.

KEY TAKEAWAY Complexity predicts scalability.

3

SESSION

60 MINUTES

LINEAR STRUCTURES

KEY TOPICS

- Arrays and dynamic arrays
- Linked lists: singly, doubly
- Stacks and queues
- Applications: undo, BFS

TEACHING FOCUS

Choose and implement linear collections.

KEY TAKEAWAY Order defines access pattern.

4

SESSION

60 MINUTES

SORTING & SEARCHING

KEY TOPICS

- Bubble, insertion, selection
- Merge sort and quick sort
- Binary search
- Hashing and hash tables

TEACHING FOCUS

Order and retrieve data efficiently.

KEY TAKEAWAY Sorting is the prerequisite for fast search.

5

SESSION

60 MINUTES

TREES & GRAPHS

KEY TOPICS

- Binary trees and BSTs
- Tree traversals
- Graph representations
- BFS/DFS and shortest path intro

TEACHING FOCUS

Model relationships and networks.

KEY TAKEAWAY Graphs model everything from circuits to cities.

6

SESSION

60 MINUTES

ALGORITHMS FOR AI & ROBOTICS

KEY TOPICS

- A* and graph search in navigation
- Algorithm selection for embedded
- Course recap and assessment
- Toward embedded systems (AI.109)

TEACHING FOCUS

Use algorithms to make robots think fast.

KEY TAKEAWAY Efficient algorithms enable real-time robots.